



Skill Exchange & Micro-Learning App

Team ID: [4] | Section: [S12] | Members: [M.Hruthika-2520090002], [K.Gayathri-2520080010],[A.Yuktha-2520090093]
| Guide: [Dr. Khaja Shareef]

Database Systems Engineering and Distributed Backend Development (25CS1302E) | KLH Bachupally Campus, CSE |
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Abstract

- CapacityConnect is a peer-to-peer skill-sharing platform that connects people who want to learn with those who can teach. It enables users to discover skills, connect with suitable peers, and exchange knowledge through two-way learning.

Problem Statement & Objectives

- Problem:** Difficulty in finding and connecting people to share and learn relevant skills through a structured platform.
- Objective 1:** Connect learners with suitable peers based on their skills and interests.
- Objective 2:** Enable users to both share skills and learn new skills.
- Objective 3:** Provide a simple platform for skill discovery, connection, and knowledge exchange.
- Scope:** Students, learners; user profiles, skill matching, connection requests, chat, and ratings; limited to peer-to-peer skill sharing.

Existing vs Proposed System

Feature	Existing System	Proposed System
Learning Model	One-way learning	Peer-to-peer learning
Skill Matching	Not specialized	Skill-based matching
Interaction	Limited interaction	Direct peer connection
Skill Sharing	Scattered	Centralized platform
Mutual Learning	Limited	Learn & teach both ways

Feasibility Study

- Technical:** Uses standard web technologies such as **HTML, CSS, JavaScript, Java, and MySQL**, which are readily available.
- Economic:** **Low-cost** development using open-source tools and free-tier hosting/database services.
- Operational:** **Simple and user-friendly** interface enables easy skill discovery, matching, connection, and communication.
- Schedule:** Can be **developed and tested within the semester timeline** using a modular implementation approach.

Results & Testing

- Key Outcome:** Users can create profiles, list skills, find suitable peers, and send connection requests.
- Testing:** CRUD operations, input validation, skill matching, and connection workflows were successfully tested.

Conclusion

- CapacityConnect connects skill seekers and skill providers through a centralized peer-to-peer platform.
- It enables skill discovery, matching, connection, and two-way knowledge sharing.

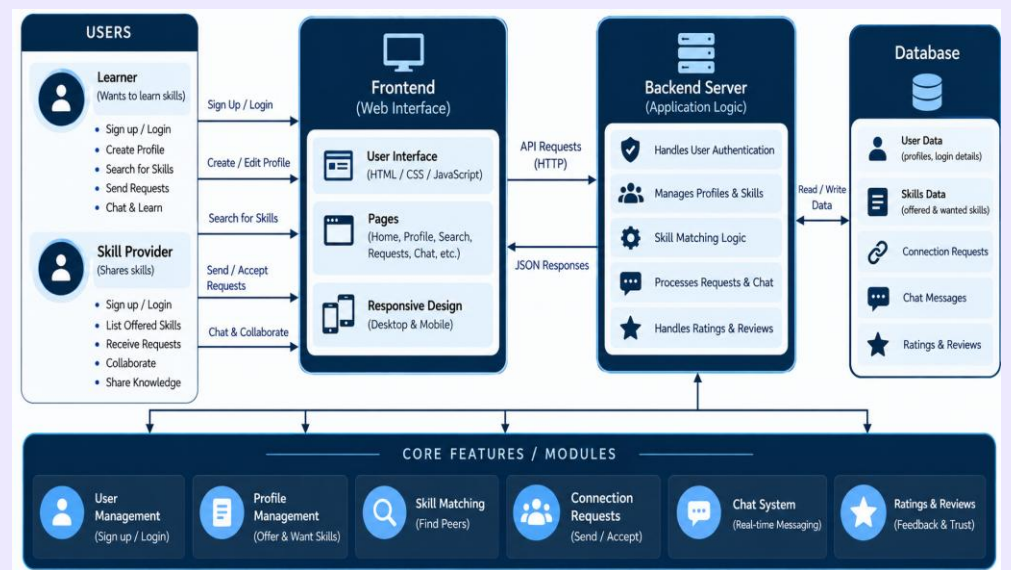
Future Scope

- AI-based skill matching and personalized recommendations
- Skill verification, badges, and advanced ratings/reviews

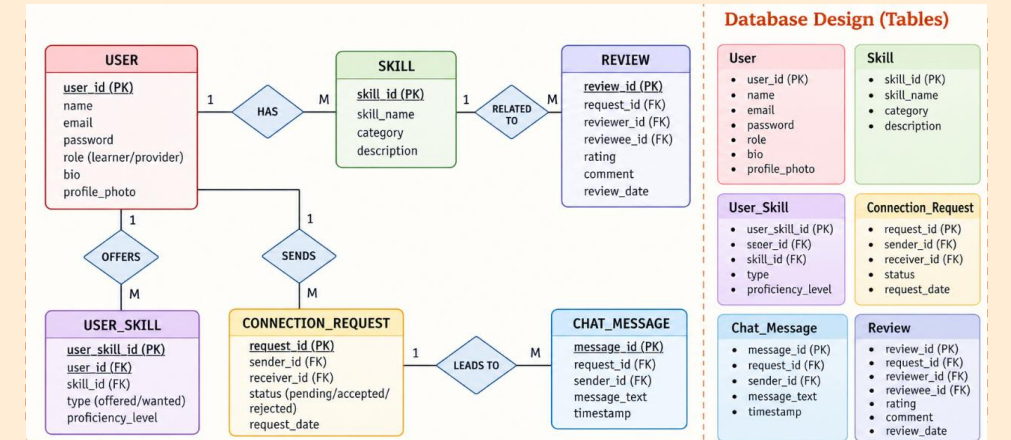
Tools & Technologies

- Front-End: [React / HTML-CSS-JS]
- Back-End: [Java/Node.js / Spring Boot] + REST APIs
- Database: [MySQL / MongoDB]
- Tools: [Git & GitHub, VS Code, Draw.io]

System Architecture



ER Diagram & Database Design



Implementation Screenshots

